

⑥ Differential Equations

→ Use separation of variables if possible.

First-order differential equations:

$$\frac{dy}{dx} + P(x) \cdot y = Q(x)$$

1st ODE

Integrating factor: ★

$$R(x) = e^{\int P(x) dx}$$

• Multiply equation to get in terms of:

$$\frac{d}{dx} (yR(x)) = Q(x)R(x)$$

2nd ODE
↓

• Particular solution: the solution when a value (initial condition) is substituted into general solution.

• General solution = Particular Integral + Comp. Function.

$$GS(x) = PI(x) + CF(x)$$

• Complementary function: auxiliary equation = 0

1) Distinct roots α, β : $Ae^{\alpha x} + Be^{\beta x}$

2) Repeated roots α : $e^{\alpha x} (A + Bx)$

3) Imaginary roots $m \pm ni$: $e^{mx} (A \cos(nx) + B \sin(nx))$

e.g. $\frac{1}{2} + \frac{\sqrt{5}i}{4} \rightarrow e^{x/2} \left(A \cos \frac{\sqrt{5}x}{4} + B \sin \frac{\sqrt{5}x}{4} \right)$

- Particular Integrals:

1) Exponents: $Ae^{\alpha x}$ $\rightarrow \alpha$ according to question

2) Polynomials: $\alpha x^2 + \beta x + \gamma$

3) Trig: $A \cos \alpha x + B \sin \alpha x$ $\rightarrow$ note both are α , same like in question

- Double differentiate and plug into the differential equation to find missing constants.

- Plug in initial conditions to find particular solution.